

University of British Columbia

MECH 464/563 EECE 442/589: Introduction to Robotics

Image Processing Homework, due Friday March 6

1. We apply erosion to the white speckled images to expand the black regions over it's white speckles, and dilation to expand the white regions over the black speckles in the black speckled regions. The best kernel shape and size is the same for both image types: size 3, shape: circle.

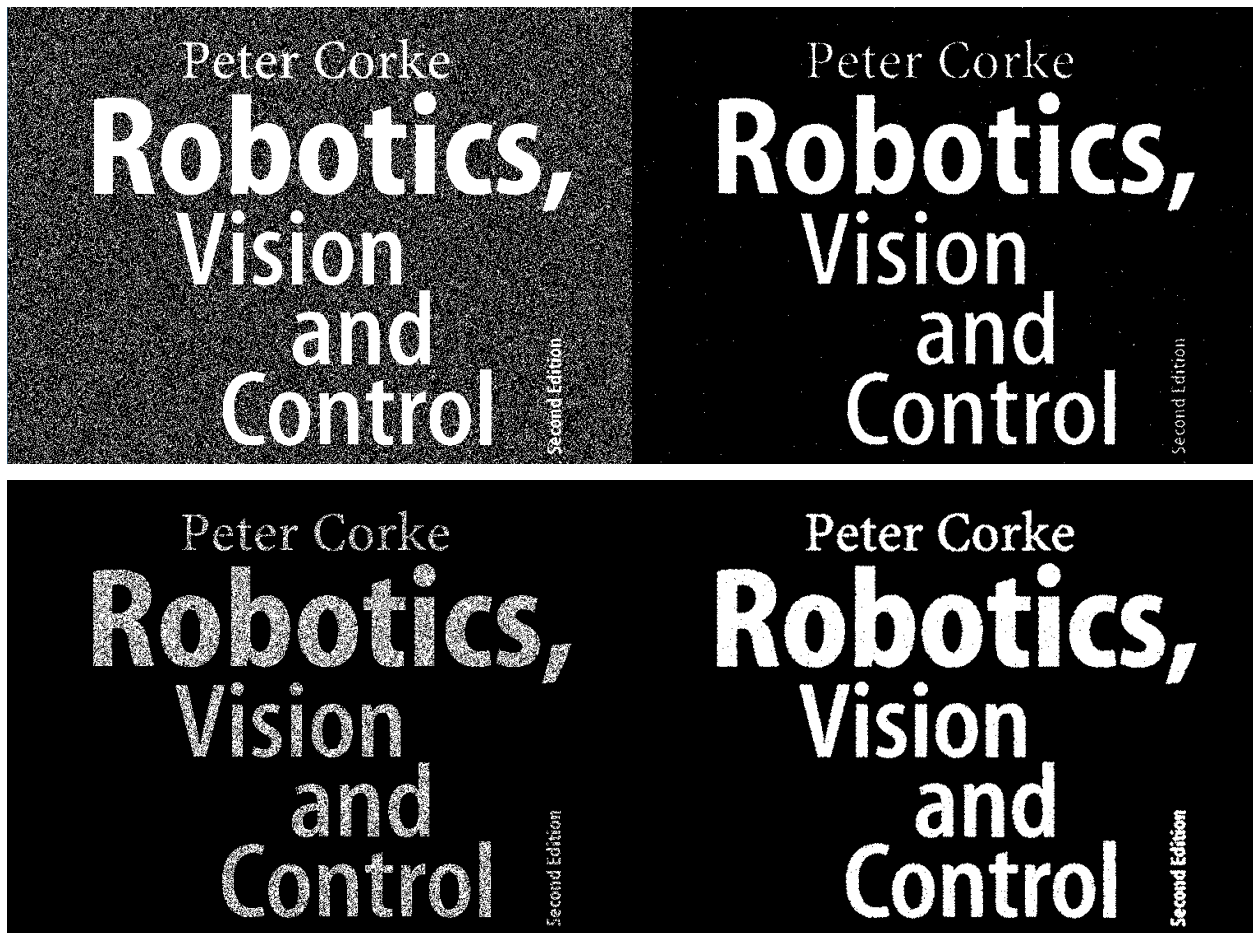


Figure 1: Best erosion (top) (SSIM 0.8965) and dialation (bottom) (SSIM 0.9039)

2. The region of interest was determined manually by trial and error, to find the rectangle precisely covering the author's name, "Peter Corke". We have the following program output:

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TASK 2: OCR / Character Recognition
=====
Author Name: Peter Corke
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3. In order to determine the colours of each cluster in lab space after applying the kmeans clustering algorithm, we hard code conditional values to determine the red, green, yellow, and blue reagions.

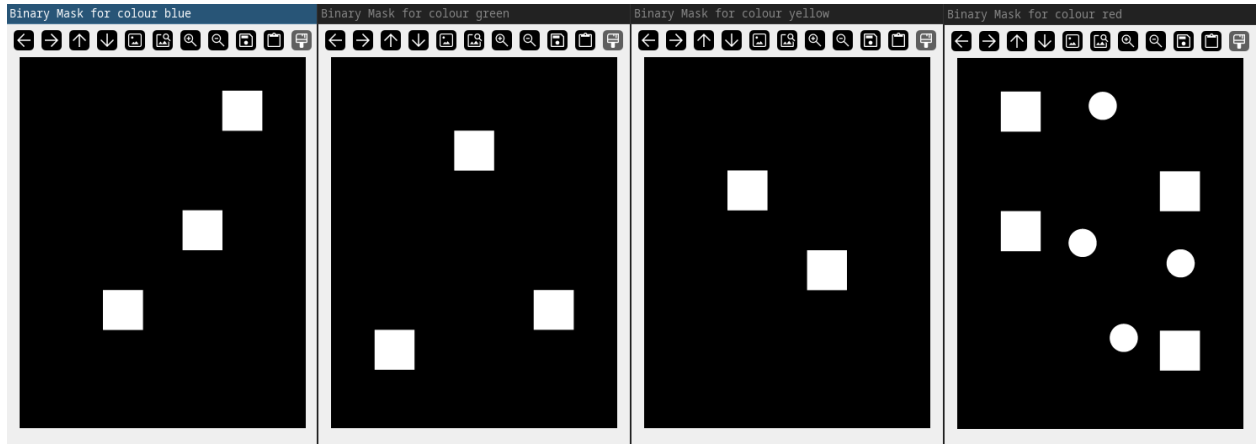


Figure 2: Binary masks for different colours, using kmeans clustering algorithm to determine regions of different colours.

4. The circles have circularity about 0.9, while the squares of circularity 0.785. Thus, we set a threshold 0.8 to mark as a circle. The images with the bounding boxes around them:

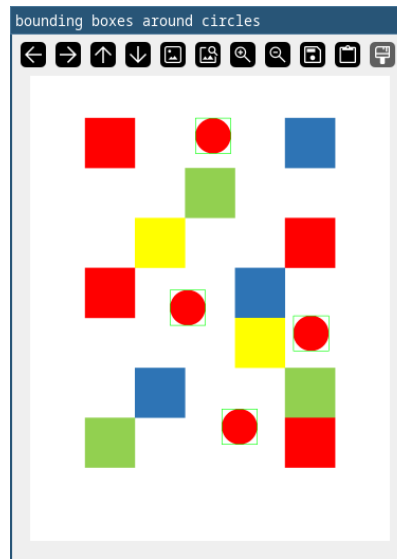


Figure 3: Bounding boxes around circles